

1. Introduction to Virtual Reality (0–10 minutes)

Definition of Virtual Reality

Virtual Reality (VR) is a computer-generated **3D simulated environment** that allows users to **experience and interact** with a virtual world in real time using specialized hardware and software.

Key Characteristics of VR:

- Computer-generated environment
- Real-time interaction
- Multi-sensory feedback (visual, audio, haptic)
- User immersion



Example:

Flight simulators, VR gaming, virtual campus tours

2. Core Fundamentals of Virtual Reality (10–30 minutes)

A. Immersion (10 minutes)

Definition:

Immersion refers to the **degree to which a user feels surrounded by the virtual environment**.

Factors affecting immersion:

- Field of View (FOV)
- Frame rate
- Display resolution
- Tracking accuracy
- Audio realism



Example:

A VR headset with 360° visuals provides higher immersion than a desktop screen.

B. Presence (10 minutes)

Definition:

Presence is the **psychological sensation of “being there”** inside the virtual environment.

Types of Presence:

- Spatial presence
- Social presence
- Self presence



Example:

Feeling afraid of heights in a VR roller-coaster simulation.

C. Interaction (10 minutes)

Definition:

Interaction is the ability of the user to **manipulate and respond to virtual objects and environments**.

Interaction Methods:

- Hand controllers
- Gesture tracking
- Eye tracking
- Voice commands



Example:

Picking up virtual objects using VR controllers.

3. Types of Virtual Reality (30–50 minutes)

A. Non-Immersive VR (10 minutes)

Description:

- Desktop-based VR
- Interaction via keyboard, mouse, joystick

Characteristics:

- Low immersion
- Low cost
- Limited interaction

Examples:

- Computer-based simulations
- 3D games on monitors

Applications:

- Education
 - Training simulations
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B. Semi-Immersive VR (10 minutes)**Description:**

- Partial immersion
- Large screens or projection systems

Characteristics:

- Moderate immersion
- Uses tracking systems

Examples:

- Flight simulators
- CAVE systems

Applications:

- Engineering training
 - Research labs
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C. Fully Immersive VR (10 minutes)**Description:**

- Complete immersion
- Uses VR headsets and sensors

Characteristics:

- High immersion and presence

- Real-time interaction

Examples:

- Meta Quest
- HTC Vive

Applications:

- Gaming
 - Medical training
 - Virtual tourism
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4. Comparison of VR Types (50–55 minutes)

Feature	Non-Immersive	Semi-Immersive	Fully Immersive
Immersion	Low	Medium	High
Hardware	Monitor	Projection	HMD
Cost	Low	Medium	High
Interaction	Limited	Moderate	Advanced

5. Recap & Discussion (55–60 minutes)

Key Takeaways:

- Immersion affects realism
 - Presence affects psychological experience
 - Interaction enables control
 - VR systems vary by immersion level
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Important Questions (Exam-Oriented)

- 1. Explain the fundamental concepts of Virtual Reality: Immersion, Presence, and Interaction.**
(8–10 marks)
- 2. Describe the different types of Virtual Reality systems with suitable examples.**
(8 marks)

3. **Compare Non-Immersive, Semi-Immersive, and Fully Immersive VR systems.**
(5 marks)